CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

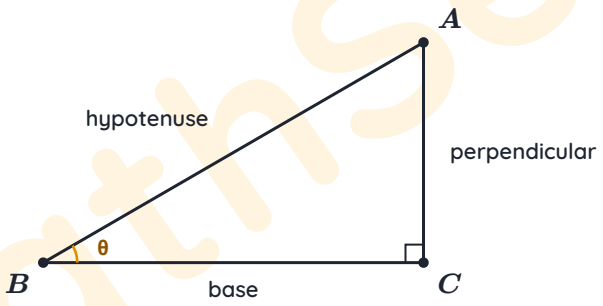
Chapter 22: Trigonometrical Ratios

Topic-wise short notes with solved examples

The six trigonometrical ratios of an acute angle in a right-angled triangle, the relations between them, and finding all the ratios when one is given.

1 Naming the sides

- In a right-angled triangle, for an acute angle θ:
- **Hypotenuse**: the side opposite the right angle.
- **Perpendicular**: the side **opposite** θ.
- **Base**: the side **adjacent** to θ (other than the hypotenuse).
- The perpendicular and the base change places when the other acute angle is used.



Right angle at C. For angle θ at B: AC is the perpendicular and BC is the base.

2 The six ratios

Ratio	Definition	Short form
sine	$\frac{\text{perpendicular}}{\text{hypotenuse}}$	$\sin\theta$
cosine	$\frac{\text{base}}{\text{hypotenuse}}$	$\cos\theta$
tangent	$\frac{\text{perpendicular}}{\text{base}}$	$\tan\theta$
cosecant	$\frac{\text{hypotenuse}}{\text{perpendicular}}$	$\operatorname{cosec}\theta$
secant	$\frac{\text{hypotenuse}}{\text{base}}$	$\sec\theta$
cotangent	$\frac{\text{base}}{\text{perpendicular}}$	$\cot\theta$

- A trigonometrical ratio depends only on the **angle**, not on the size of the triangle.
- $\sin\theta$ is one symbol. It does not mean $\sin\times\theta$. $(\sin\theta)^2$ is written $\sin^2\theta$.

- Since the hypotenuse is the longest side, $\sin\theta$ and $\cos\theta$ are less than 1, and $\operatorname{cosec}\theta$ and $\sec\theta$ are greater than 1, for an acute angle.

Example 2.1 In triangle ABC, $\angle B = 90^\circ$, $AB = 5$ cm and $BC = 12$ cm. Find $\sin A$, $\cos A$ and $\tan C$.

SOLUTION

$AC = \sqrt{25 + 144} = 13$ cm. For angle A: perpendicular $BC = 12$, base $AB = 5$.

For angle C: perpendicular $AB = 5$, base $BC = 12$.

$\sin A = \frac{12}{13}$, $\cos A = \frac{5}{13}$, $\tan C = \frac{5}{12}$

3 Relations between the ratios

- **Reciprocal relations**: $\operatorname{cosec}\theta = \frac{1}{\sin\theta}$, $\sec\theta = \frac{1}{\cos\theta}$, $\cot\theta = \frac{1}{\tan\theta}$.
- **Quotient relations**: $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\cot\theta = \frac{\cos\theta}{\sin\theta}$.
- **Square relations** (from Pythagoras theorem):

$$\sin^2\theta + \cos^2\theta = 1, \qquad 1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

Example 3.1 If $\sin\theta = \frac{3}{5}$, find $\cos\theta$ using a square relation.

SOLUTION

$\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$

$\cos\theta = \frac{4}{5}$

4 Given one ratio, find the others

- Draw a right triangle and mark two sides from the given ratio, using a common multiple k.
- Find the third side by Pythagoras theorem.
- Write the required ratios. The k cancels.

Example 4.1 If $\tan\theta = \frac{8}{15}$, find $\sin\theta$ and $\sec\theta$.

SOLUTION

Perpendicular $8k$, base $15k$, hypotenuse $\sqrt{64k^2 + 225k^2} = 17k$.

$\sin\theta = \frac{8}{17}$, $\sec\theta = \frac{17}{15}$

Example 4.2 If $\cos A = \frac{7}{25}$, find $\tan A + \cot A$.

SOLUTION

Base $7k$, hypotenuse $25k$, perpendicular $24k$. $\tan A = \frac{24}{7}$, $\cot A = \frac{7}{24}$.

$$\frac{24}{7} + \frac{7}{24} = \frac{576 + 49}{168}$$

$$\frac{625}{168}$$

Example 4.3 If $\operatorname{cosec}\theta = \sqrt{5}$, find $\cot\theta - \cos\theta$.

SOLUTION

Hypotenuse $\sqrt{5}k$, perpendicular k, base $2k$. $\cot\theta = 2$, $\cos\theta = \frac{2}{\sqrt{5}}$.

$$2 - \frac{2}{\sqrt{5}} = \frac{2(5 - \sqrt{5})}{5}$$

Example 4.4 If $5\sin\theta = 3$, find the value of $\frac{\sec\theta - \tan\theta}{\sec\theta + \tan\theta}$.

SOLUTION

$\sin\theta = \frac{3}{5}$: perpendicular 3, hypotenuse 5, base 4. $\sec\theta = \frac{5}{4}$, $\tan\theta = \frac{3}{4}$.

$$\frac{\frac{5}{4} - \frac{3}{4}}{\frac{5}{4} + \frac{3}{4}} = \frac{\frac{1}{2}}{2}$$

$$\frac{1}{4}$$

5 Using the ratios to find sides

- In a right triangle, if one side and a ratio are known, the other sides follow.

Example 5.1 In triangle PQR, $\angle Q = 90^\circ$, $PR = 20$ cm and $\sin R = \frac{3}{5}$. Find PQ and QR.

SOLUTION

$\sin R = \frac{PQ}{PR}$, so $PQ = 20 \times \frac{3}{5} = 12$ cm. $QR = \sqrt{400 - 144}$.

$PQ = 12$ cm, $QR = 16$ cm

Example 5.2 In a rhombus of side 10 cm, a diagonal makes an angle θ with a side, where $\cos\theta = \frac{4}{5}$. Find the diagonals.

SOLUTION

The diagonals bisect each other at right angles. Half of one diagonal $= 10\cos\theta = 8$ cm; half of the other $= 10\sin\theta = 6$ cm.

16 cm and 12 cm

6 Expressions in sine and cosine

- To evaluate an expression in $\sin\theta$ and $\cos\theta$ when $\tan\theta$ is known, **divide the numerator and the denominator by** $\cos\theta$.

Example 6.1 If $\tan\theta = \frac{4}{3}$, find $\frac{3\sin\theta + 2\cos\theta}{3\sin\theta - 2\cos\theta}$.

SOLUTION

Dividing by $\cos\theta$: $\frac{3\tan\theta + 2}{3\tan\theta - 2} = \frac{4+2}{4-2}$

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Example 6.2 If $\cot\theta = \frac{1}{\sqrt{3}}$, show that $\frac{1-\cos^2\theta}{2-\sin^2\theta} = \frac{3}{5}$.

SOLUTION

Base 1, perpendicular $\sqrt{3}$, hypotenuse 2: $\cos\theta = \frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$.

$\frac{1-\frac{1}{4}}{2-\frac{3}{4}} = \frac{\frac{3}{4}}{\frac{5}{4}} = \frac{3}{5}$

7 Common mistakes

- Using the same perpendicular and base for both acute angles. They swap.
- Writing $\sin^2\theta$ as $\sin\theta^2$ or $\sin 2\theta$.
- Taking $\operatorname{cosec}\theta$ as the reciprocal of $\cos\theta$. It is the reciprocal of $\sin\theta$.
- Cancelling "sin" as if it were a number: $\frac{\sin A}{\sin B}$ is not $\frac{A}{B}$.
- Forgetting the square root when finding the third side.
- Getting a value of $\sin\theta$ or $\cos\theta$ greater than 1 and not noticing the error.